

Reader Report: Week 05

Bayesian Invariant Prediction & Optimization-Based ICP

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Summary

Wu and Blei (2025) introduce **Bayesian Invariant Prediction** (BIP), casting ICP as latent-variable inference. A binary selector $z \in \{0, 1\}^p$ indexes invariant features; the posterior $p(z \mid \mathcal{D})$ is proportional to likelihood ratios between pooled and per-environment conditionals. They prove consistency (Thm 1) and exponential contraction at rate $O(R \cdot e^{-\kappa n E \mu_{\min}})$ (Thm 2), where $\mu_{\min}$ captures environment heterogeneity. For scalability, **VI-BIP** uses mean-field variational inference with the U2G gradient estimator; it identifies invariant genes in a 6,170-gene perturbation dataset where competing methods falter.

Fan et al. (2024) propose **NegDRO**, a continuous minimax formulation: $\min_b \max_{w \in U(\gamma)} \sum_e w_e \mathbb{E}[Y^e - b^\top X^e]^2$, where $U(\gamma)$ allows *negative weights* to enforce risk invariance. Under additive interventions, every stationary point is $O((1+\gamma|E|)^{-1} + T^{-1/4} + n^{-1/4})$ -close to β^* (Thm 5), enabling gradient descent in *polynomial* time rather than 2^p enumeration.

Contrast: BIP gives full posterior uncertainty over invariant sets; NegDRO trades distributional characterization for polynomial-time convergence with provable rates.

Critique & Project Connection

For my *Predicting NIMBYism Causally* project (~ 4 temporal environments, $p \approx 50$ features), BIP's multi-modal posterior honestly quantifies which invariant sets remain ambiguous with limited environments. NegDRO's polynomial-time guarantee and tolerance of hidden confounding (e.g., unmeasured wealth confounding school quality and opposition) better suit iterative model development. A hybrid strategy could use NegDRO for fast screening, then BIP for uncertainty quantification on a reduced feature set. **Limitation:** both assume linear/squared-loss models; extending to binary outcomes ($Y \in \{0, 1\}$) remains future work.

Questions

1. BIP's contraction depends on $\mu_{\min}$ (KL divergence between local and pooled conditionals). How should one validate sufficient environment heterogeneity *before* running BIP?
2. NegDRO achieves $n^{-1/4}$ sample complexity. Could sparsity of β^* yield faster rates toward the $n^{-1/2}$ parametric rate?

Project Update: Predicting NIMBYism Causally

1. What is the problem?

We seek the causal drivers of zoning opposition (NIMBYism) in Austin, TX. Following Peters et al. (2016), we adopt invariant causal prediction (ICP): find S^* such that $P_e(Y \mid X_{S^*}) = P(Y \mid X_{S^*})$ across all environments e . Features stable across regimes are identified as causal; those whose predictive power breaks are discarded as spurious.

The key to identification is environment diversity—each regime change stress-tests predictors, eliminating spurious ones. Over 2019–2024, Austin experienced the Adler-to-Watson mayoral transition (Jan 2023), parking-reform elimination (Nov 2023), the HOME Initiative (Feb 2024), and council-seat turnovers. These yield *temporal* environments (citywide reforms) and *geographic* ones (council-district turnovers; parcels outside Austin city limits as a natural control). Each event intervenes on a subset of covariates while preserving the causal mechanism (Bühlmann, 2020, assumption B), constituting **limited additive interventions** (Fan et al., 2024, §5). With ~ 50 features, Peters et al.’s intersection $\hat{S}(E) \subseteq S^*$ tightens monotonically with each additional environment (Peters et al., 2016, Thm 1); Fan et al.’s Condition 4 extends this to partial interventions, and BIP’s multi-modal posterior quantifies remaining ambiguity.

2. Mathematical Formulation

Setup. $\mathcal{E} = \{e_1, \dots, e_K\}$: environments from temporal breakpoints (HOME, parking reform, mayoral transition) crossed with geographic variation (council-district turnovers, city-limits boundary). In each e observe $(X_i^e, Y_i^e)_{i=1}^{n_e}$ with $X^e \in \mathbb{R}^p$ (valuations, lot dimensions, land-use class, tax exemptions) and $Y^e \in \{0, 1\}$ (protest signed). Covariates split: *intervened* S_L (regulatory/market features) vs. *untouched* S_L^c (demographics, schools, transit).

Causal model. We adopt the linear SEM with limited additive interventions because it matches our data—reforms intervene on some covariates but not others—and both methods provide their strongest guarantees under this class:

$$Y^e = (\beta^*)^\top X_{S^*}^e + \varepsilon_Y^e, \quad \varepsilon_X^e \stackrel{d}{=} \eta_X + (0_{S_L^c}, \delta_{S_L}^e)^\top, \quad \mathbb{E}[\eta(\delta^e)^\top] = 0.$$

$S^* \subseteq [p]$: true causal parents; ε_Y^e invariant; only $\delta_{S_L}^e$ varies. Mechanism $Y \mid X_{S^*}$ stable; reforms shift $P(X^e)$ only.

Objective. Find S^*, β^* : $\mathbb{E}[(Y^e - (\beta^*)^\top X^e)^2] = \sigma_Y^2$ for all e (risk invariance).

NegDRO (Fan et al., 2024). $\hat{\beta}^\gamma \in \arg \min_b \max_{w \in U(\gamma)} \sum_e w_e \mathbb{E}[(Y^e - b^\top X^e)^2]$; $U(\gamma)$ allows negative weights. Under Condition 4, recovers β^* with limited interventions.

BIP (Wu and Blei, 2025). Latent $z \in \{0, 1\}^p$; posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z) / \hat{p}_e(Y_i^e \mid X_{ei}^z)$ concentrates on z^* as $n, K \rightarrow \infty$.

References

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